

2-WAY CROSSOVER

ScanSpeak 18W8531G00

ScanSpeak D3004-660000

Complete Simulation Graphs & Crossover Design

Contents

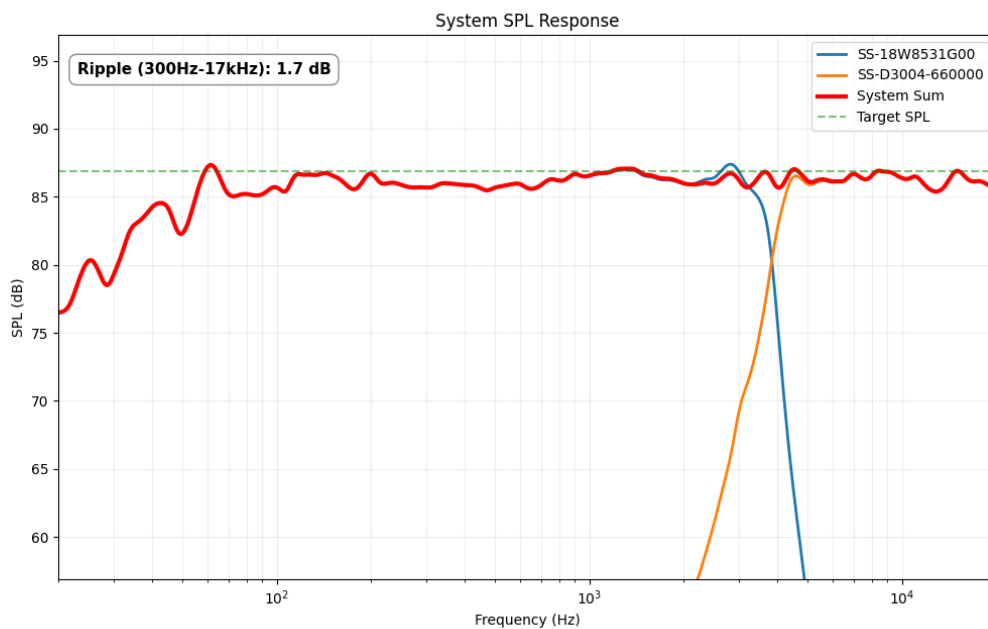
1	Acoustic Simulations	2
1.1	On-Axis SPL Response	2
1.2	Off-Axis Response	3
1.3	Directivity Heatmap	3
1.4	Impedance Response	4
2	Bill of Materials (BOM)	5
3	Crossover Design	6
3.1	Electrical Schematic	6
3.2	Baffle Geometry	6

1

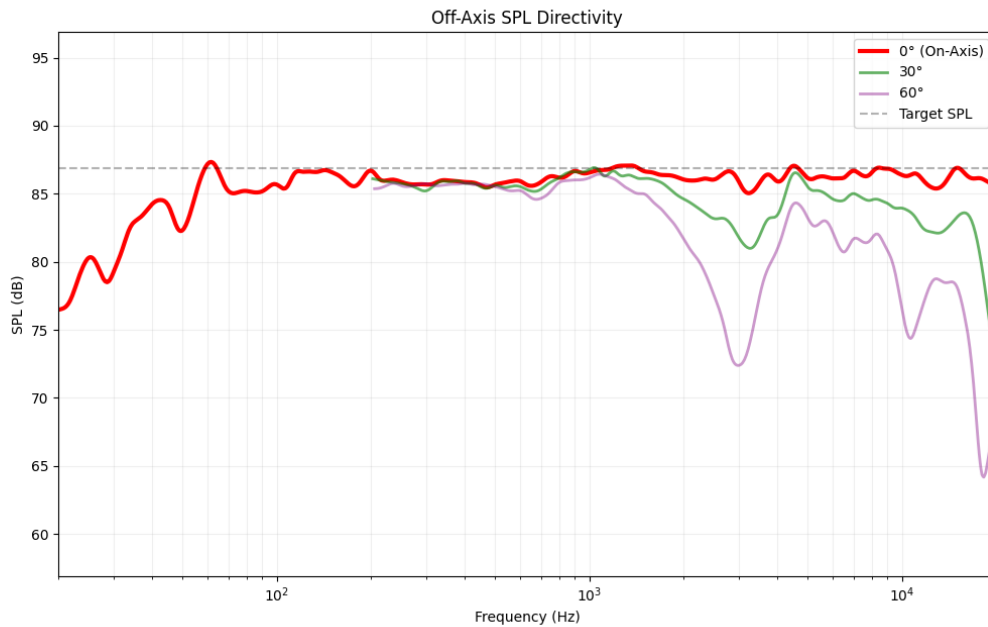
Acoustic Simulations

Simulation graphs showing the frequency response of the drivers combined with the crossover. For more information on how to interpret these graphs, please refer to the beginner guide provided with your order.

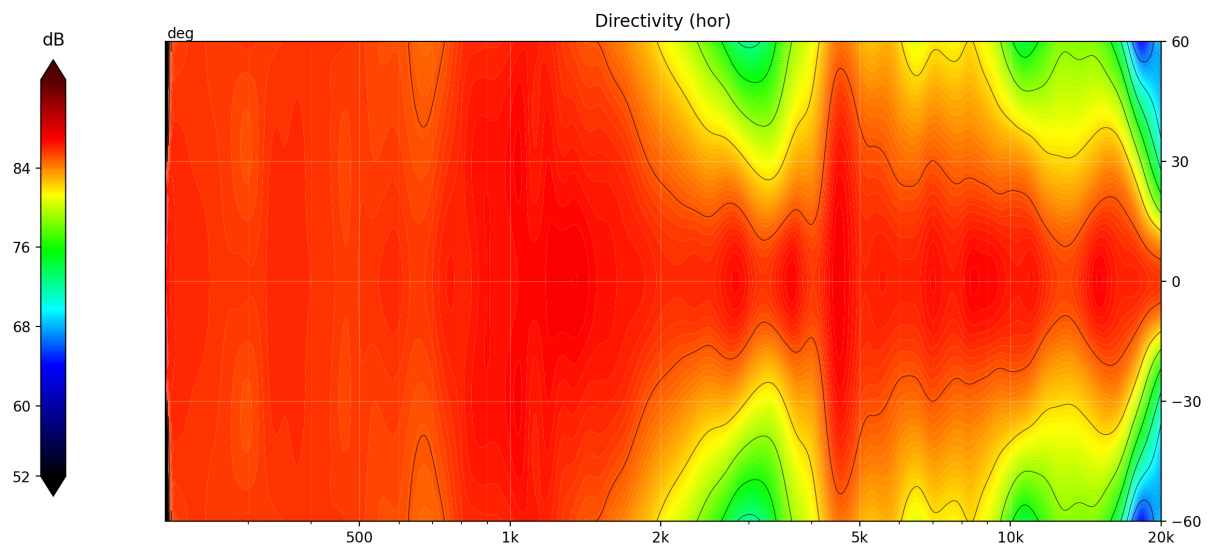
1.1 On-Axis SPL Response



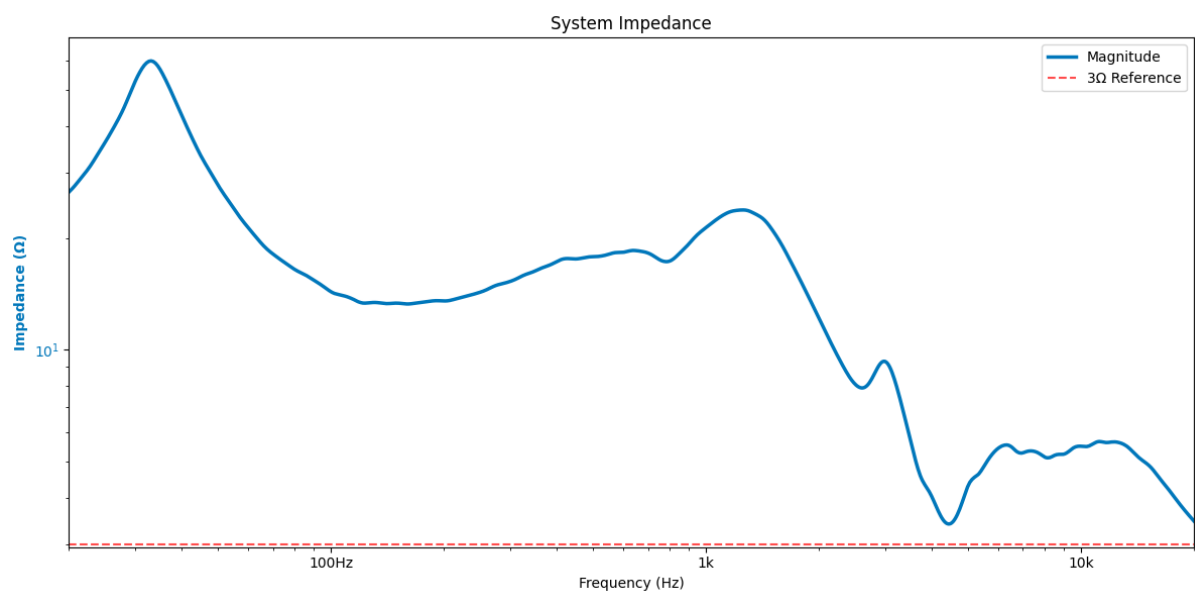
1.2 Off-Axis Response



1.3 Directivity Heatmap



1.4 Impedance Response



2

Bill of Materials (BOM)

The following table lists all the components required to build one crossover. For a stereo pair, please double the quantities. Note that the provided prices and links are for reference only and may vary over time. Always check the supplier's website for the most up-to-date information.

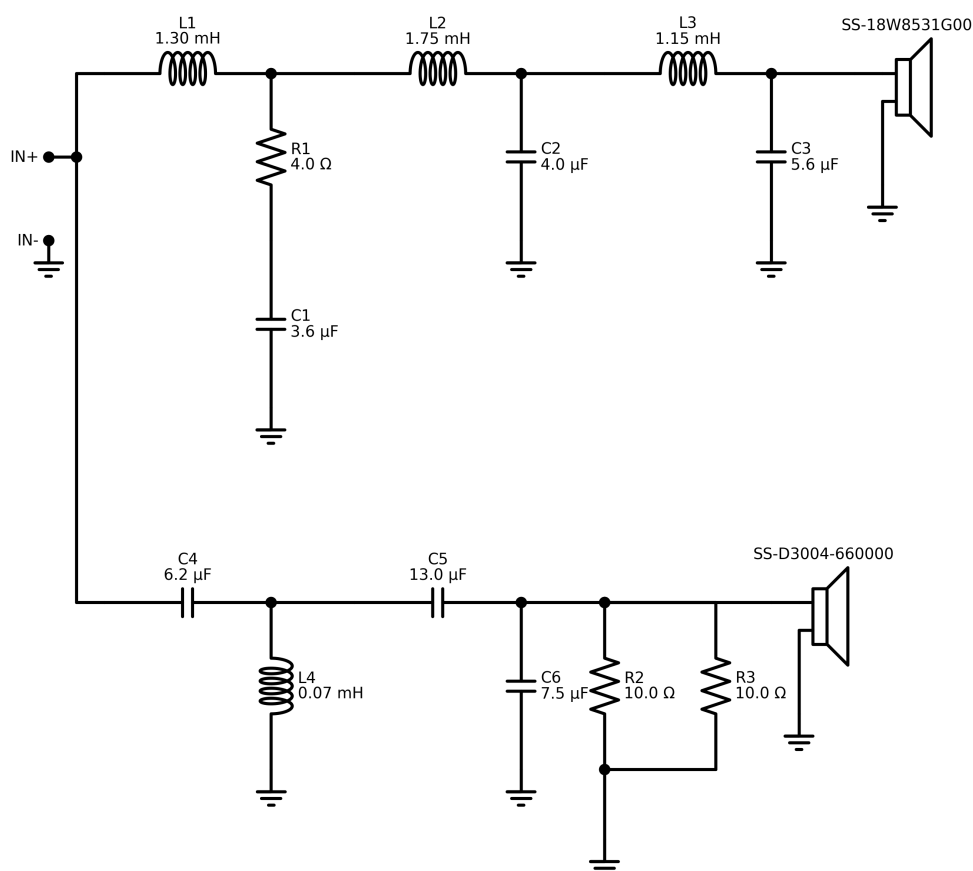
ID	Component	Value	Price (\$/€)	Buy Link
L1	Inductor	1.3 mH	13.46	Link
R1	Resistor	4.0 Ω	1.25	Link
C1	Capacitor	3.6 μ F	2.44	Link
L2	Inductor	1.75 mH	9.95	Link
C2	Capacitor	4.0 μ F	0.89	Link
L3	Inductor	1.15 mH	13.95	Link
C3	Capacitor	5.6 μ F	0.99	Link
C4	Capacitor	6.2 μ F	1.99	Link
L4	Inductor	0.07 mH	18.45	Link
C5	Capacitor	13.0 μ F	10.25	Link
C6	Capacitor	7.5 μ F	2.29	Link
R2	Resistor	10.0 Ω	0.99	Link
R3	Resistor	10.0 Ω	0.99	Link
Estimated Total			77.89	

3

Crossover Design

This section provides the electrical schematic required to build the crossover and achieve the acoustic results presented earlier. If you need help understanding this diagram, please refer to the beginner guide.

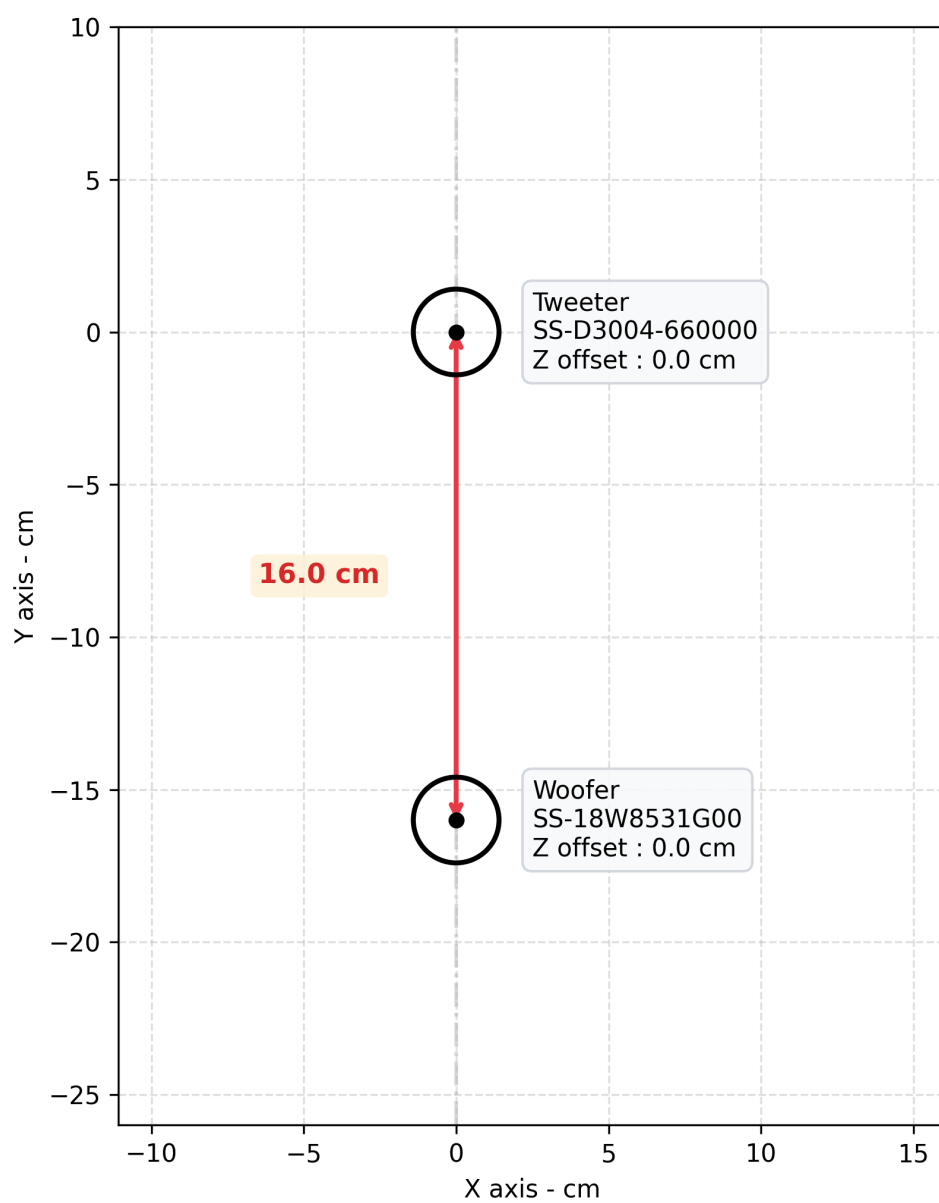
3.1 Electrical Schematic



3.2 Baffle Geometry

All figures are computed using the following baffle layout. We recommend using this layout for optimal acoustic performance, but feel free to experiment with different placements if you have specific constraints or preferences. Just keep in mind that drastically changing the geometry may affect the frequency response and directivity of your speakers due to phase shifts.

Baffle Geometry



THANK YOU

FOR YOUR SUPPORT

I hope this guide helps you build the crossover for your DIY audio project.

If you have any questions, run into issues, or just want to share a picture of your finished build, please feel free to reach out via Etsy messages!

Happy building,

Audio440



www.etsy.com/shop/Audio440